Practice test: semester A

Test Content

Please attempt all the questions below. As well as the (largely numerical) answers in the response boxes ***please ensure that you upload a supporting file with your working, extra information or answers to sub questions and code*** to allow me to give you partial credit for questions.

You have 1.5 hours to attempt the questions (unless your PASS plan gives you more time). I will allow a little extra time to format and upload your supporting file but please try to finish within the time allotted.

Question 1

15 Points

Perform linear least squares regression on the following data:

x_i y_i

0.000000 85.358681
1.428571 14.165804
2.857143 -26.825715
4.285714 -4.997952
5.714286 -94.922415
7.142857 -164.033960
8.571429 -180.876888
10.00000 -206.688112

Give the correlation coefficient for the data correct to 3 decimal places.

Include a plot of your model along with the original data in your supporting information.

Add your answer

Integer, decimal or E notation allowed

Question 2

5 Points

Which of the following statements about the assumptions behind linear regression is FALSE

(A) The y values are randomly distributed with the same variance.

(B) The y values of the data are normally distributed

(C) All the data points are measured exactly.

(D) Each x value is precisely measured.

Question 3

10 Points

After fitting a model to data, what checks would you perform to confirm that this has been done correctly?

Use the editor to format your answer

Question 4

15 Points

Find using the sum of the solutions, $x_0 + x_1 + x_2$, of the following set of equations: $\begin{bmatrix} 9.88147746930662 & 9.446294330943026 & 6.867343600899789 \\ 3.3013023006376 & 8.866849207559063 & 2.5855821434029 \\ 7.897613140999123 & 2.5810421630855136 & 9.843121360475344 \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1.7147395700538763 \\ 3.82744793002092 \\ 8.883242030765913 \end{bmatrix}$

Add your answer

Integer, decimal or E notation allowed

Question 5

15 Points

Using Newton's method or the Secant method, find the root(s) of
$x^2 + 2.5x = 4.6875$.
Give your numerical answer here.

Explain the Secant method and explain how it is related to Newton's method in your uploaded supporting file.

Add your answer

Integer, decimal or E notation allowed

Question 6

20 Points

Solve the following differential equation using finite differences and solving the resulting set of linear equations.

$$q\frac{\mathrm{d}^2H(g)}{\mathrm{d}g^2} + kH(g) = b$$

where $g \in [0,5]$.

Use the boundary conditions, $H_L = 5$, $H_R = 5$, and set $q = -3.0$, $k = 0$, $b = 6$.

Give the maximum value of H (g) as your numerical answer below.

In your supporting file:

Plot the function $H(g)$ in the interval $[0,5]$ using $10,50,100,200$ points when discretizing the independent variable.

If $k = 5.538013$ how would you tackle the problem and how would the solution change?

Add your answer

Integer, decimal or E notation allowed

Question 7

20 Points

The cosine coefficients in a Fourier series expansion of a periodic function on the interval $[-l,l)$ can be defined as

$a_m = \frac{1}{l}\int_{-l}^{l} \phi_m(x) f(x) \mathrm{d}x$, where $\phi_m(x) = \cos(m\pi x / l)$

which can be approximated numerically as

$a_m = \frac{1}{l}\sum_{i=0}^{n} \phi_m(x_i) f(x_i) dx$

on a regular grid of values, x_i, of size N.

Calculate numerically a_1 (the coefficient of $\cos(2\pi x)$) in the Fourier series expansion of $f(x) = 4(x-1/2)^2$ defined on $[0,1)$.

In your supporting file, explain the numerical approximation to obtain a_m.

Add your answer

Integer, decimal or E notation allowed

Additional content

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